

premium were less than the intrinsic value, a trader could buy and exercise the option and immediately offset the resulting market position, thereby realizing a net gain (assuming that the trader covers at least transaction costs).

Options that have intrinsic value (i.e., calls with strike prices below the market price and puts with strike prices above the market price) are said to be *in-the-money*. Options that have no intrinsic value are called *out-of-the-money* options. Options with a strike price closest to the market price are called *at-the-money* options.

An out-of-the-money option, which by definition has an intrinsic value equal to zero, will still have some value because of the possibility that the market price will move beyond the strike price prior to the expiration date. An in-the-money option will have a value greater than the intrinsic value because, if priced at the intrinsic value, a position in the option would always be preferred to a position in the underlying market. Why? Because both the option and the market position would then gain equally in the event of a favorable price movement, but the option's maximum loss would be limited. The portion of the premium that exceeds the intrinsic value is called the *time value*.

The three most important factors that influence an option's time value are:

1. *Relationship between the strike and market price*—Deeply out-of-the-money options will have little time value since it is unlikely that the market price will move to the strike price—or beyond—prior to expiration. Deeply in-the-money options have little time value, because these options offer positions very similar to the underlying market—both will gain and lose equivalent amounts for all but an extremely adverse price move. In other words, for a deeply in-the-money option, the fact that risk is limited is not worth very much, because the strike price is so far from the prevailing market price.
2. *Time remaining until expiration*—The more time remaining until expiration, the greater the value of the option. This is true because a longer life span increases the probability of the intrinsic value increasing by any specified amount prior to expiration.
3. *Volatility*—Time value will vary directly with the estimated *volatility* [a measure of the degree of price variability] of the underlying market